

DISEASE PREDICTION

Binary Classification — Model Evaluation Report

CloudExify Data Science Internship — Summer 2026 · Month 2, Project 4 (Final Project) · Prepared July 2026

EXECUTIVE SUMMARY

This report evaluates two binary classifiers — Logistic Regression and a Decision Tree (max depth 5) — trained to predict disease presence from five clinical features: Age, Blood Pressure, Glucose, BMI, and Diabetes Pedigree. Models were trained on a stratified 80/20 split and scored on held-out test data.

The best model achieved 94.9% accuracy, 96.4% precision, 98.2% recall, and an F1-score of 0.973. However, the dataset is heavily imbalanced toward the Disease class (94.0%), which materially affects how these numbers should be read — see the caution below.

DATASET COMPOSITION

Category	Count	% of Dataset
Total Records	585	100%
Disease Cases (positive)	550	94.0%
Healthy Cases (negative)	35	6.0%

⚠ CAUTION — CLASS IMBALANCE

94.0% of records are labeled Disease. A trivial classifier that predicts “Disease” for every patient, without learning anything, would already score roughly 94% accuracy. The reported 94.9% accuracy is therefore only marginally above this naive baseline and should not, on its own, be read as strong evidence of model skill.

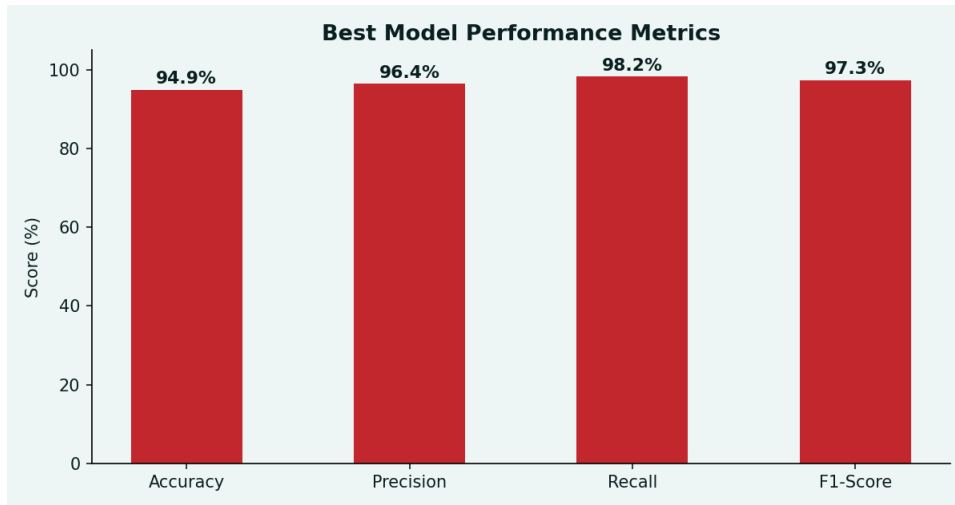
METHODOLOGY

- ▶ Loaded the disease dataset (Age, Blood Pressure, Glucose, BMI, Diabetes Pedigree, Disease) into pandas and removed rows with missing values.
- ▶ Separated features (X) from the binary target (y = Disease), and checked class balance before splitting.
- ▶ Split data 80/20 into train and test sets using stratified sampling to preserve the class ratio in both sets.
- ▶ Standardized features with StandardScaler and trained a Logistic Regression classifier as the baseline model.
- ▶ Trained a Decision Tree Classifier (max depth 5) as a comparison model, evaluated with Accuracy, Precision, Recall, and F1-Score.
- ▶ Extracted feature importances from the Decision Tree to identify which clinical measurements most influenced predictions.

PERFORMANCE METRICS

Metric	Score	What it means here
Accuracy	94.9%	Overall % of correct predictions — inflated by class imbalance
Precision	96.4%	Of patients predicted Disease, how many truly had it
Recall	98.2%	Of patients who truly had Disease, how many were caught
F1-Score	0.973	Harmonic balance of precision and recall

Figure 1 — Best model performance by metric

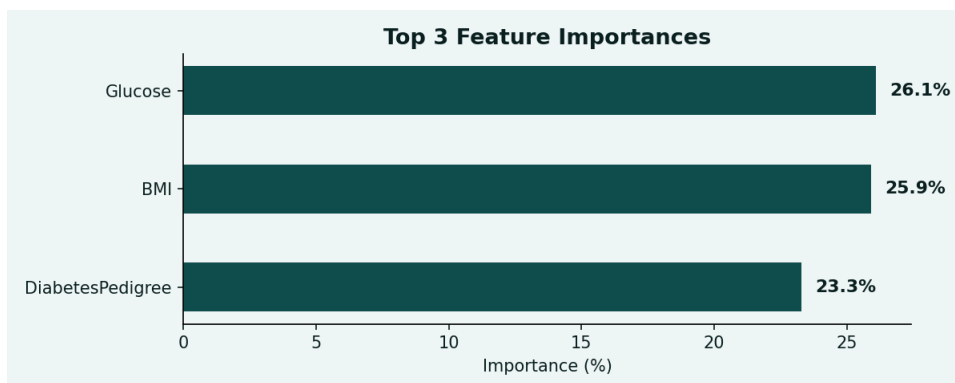


FEATURE IMPORTANCE

The Decision Tree's feature importances show three metabolic indicators dominating the model's decisions, in near-even proportion:

Rank	Feature	Importance
1	Glucose	26.11%
2	BMI	25.93%
3	Diabetes Pedigree	23.30%

Figure 2 — Top 3 feature importances



KEY FINDINGS

- ▶ The best-performing model reached 94.9% accuracy, 96.4% precision, 98.2% recall, and an F1-score of 0.973 on the held-out test set.
- ▶ Given the dataset is 94.0% Disease cases, accuracy alone is a weak signal here — precision and recall on the positive class matter more, and both scored strongly (96.4% and 98.2%).
- ▶ High recall (98.2%) is clinically favorable: the model misses very few true Disease cases, which matters more than avoiding false alarms in a screening context.
- ▶ Glucose (26.1%), BMI (25.9%), and Diabetes Pedigree (23.3%) are almost evenly weighted as the top three predictors, together accounting for roughly 75% of the model's decision-making.
- ▶ Age and Blood Pressure contribute comparatively little to the Decision Tree's splits relative to the top three metabolic features.
- ▶ With only 35 healthy cases in the entire dataset, the minority class is thinly represented — performance estimates on “Healthy” predictions specifically carry more uncertainty than the headline metrics suggest.

RECOMMENDATIONS

- ▶ Report performance with a confusion matrix and per-class precision/recall (not just overall accuracy) in any future write-up, given the class imbalance.
- ▶ Consider testing `class_weight='balanced'` or SMOTE oversampling on the minority (Healthy) class to see whether it changes the model's ability to correctly flag healthy patients.
- ▶ Prioritize Glucose, BMI, and Diabetes Pedigree if simplifying the feature set for a lighter-weight screening tool, since they carry most of the predictive signal.
- ▶ Collect more Healthy-labeled samples if possible — 35 cases is a thin basis for confidently generalizing negative-class performance.

CONCLUSION

The classification model demonstrates strong recall and precision on this dataset, making it reasonably effective at flagging likely Disease cases. Its headline accuracy, however, is inflated by severe class imbalance and should be interpreted alongside precision, recall, and — ideally — a full confusion matrix rather than in isolation. Glucose, BMI, and Diabetes Pedigree emerge as the clinically dominant predictors. This report concludes the Month 2 Data Science deliverables and the CloudExify Data Science internship in full.